

Geon-Tae Park

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 Scholar  ORCID

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INTERESTS Machine learning-guided inverse design of optical and photonic materials and structures for energy, sensing, and cross-disciplinary applications.

EDUCATION **Kyung Hee University (KHU)** **Yongin, Korea**
MS in Physics (GPA: 4.3 / 4.3) (Advisor: Prof. Sun-Kyung Kim) *Mar. 2024 – Feb. 2026*
BS in Applied Physics (GPA: 3.85 / 4.3) *Mar. 2018 – Feb. 2024*

- PUBLICATIONS**
- [1] **Geon-Tae Park**, J. He, E. Lee, S.-K. Kim, R. Hu (2026). “Interpretable-Machine-Learning-Enabled Discretized Antireflective Multilayer Design Outstands Traditional Continuous Graded-Index Coatings,” *Applied Optics*, 65 (12), 4018–4030.  
 - [2] **Geon-Tae Park**, J.-W. Cho, H. R. Kim, Q. Gan, Y. M. Song, S.-K. Kim (2025). “Microstructured Optical Coatings for Climate Crisis Mitigation,” *Applied Physics Reviews*, 12 (3), 031332.  
 - [3] J.-H. Kim, **Geon-Tae Park**, R. Kang, B. Lee, S.-K. Kim (2025). “Geometric Design for Light Trapping in Bifacial Thin-Film Photovoltaics,” *IEEE Journal of Selected Topics in Quantum Electronics*, 31, 4000107.  
 - [4] **Geon-Tae Park**, R. Kang, B. Lee, S.-K. Kim (2024). “Binary-Optimization-Based Multilayers and Their Practical Applications,” *Current Optics and Photonics*, 8 (6), 545–561.  
 - [5] **Geon-Tae Park**, J.-H. Kim, S. Lee, D. I. Kim, K.-S. An, E. Lee, S. Yim, S.-K. Kim (2024). “Conformal Antireflective Multilayers for High-Numerical-Aperture Deep-Ultraviolet Lenses,” *Advanced Optical Materials*, 12 (26), 2401040.  
 - [6] J.-S. Yu, S. Jung, J.-W. Cho, **Geon-Tae Park**, Mikhail K., S.-K. Kim, E. Lee (2024). “Ultrathin Ge-YF₃ Antireflective Coating with 0.5% Reflectivity on High-Index Substrate for Long-Wavelength Infrared Cameras,” *Nanophotonics*, 13 (21), 4067–4078.  

- CONFERENCE PRESENTATIONS**
- [1] **Geon-Tae Park**, E. Lee, S.-K. Kim (Jul. 2025). “Benchmarking Binary Optimization and Graded-Index Approaches for Deep-Ultraviolet Antireflective Coatings,” *Optics and Photonics Congress*, Jeju, Korea.
 - [2] **Geon-Tae Park**, S.-K. Kim (Feb. 2025). “Leveraging Binary Optimization for Robust Multilayer Design,” *Winter Academic Conference*, Busan, Korea.
 - [3] J.-S. Yu, S. Jung, J.-H. Kim, **Geon-Tae Park**, E. Lee, S.-K. Kim (Apr. 2024). “Broadband Wide-Angle Antireflective Coatings for Long-Wave Infrared Cameras,” *PIERS*, Chengdu, China.
 - [4] **Geon-Tae Park**, J.-H. Kim, S. Lee, D. I. Kim, K.-S. An, E. Lee, S. Yim, S.-K. Kim (Jul. 2024). “Conformal Antireflective Coatings for High-Numerical-Aperture Lenses,” *Optics and Photonics Congress*, Jeju, Korea.
 - [5] **Geon-Tae Park**, J.-H. Kim, S. Yim, E. Lee, S.-K. Kim (Aug. 2023). “Evaluation of Wide-Angle Antireflective Multilayers with Varying Refractive Index Contrast,” *Optics and Photonics Congress*, Jeju, Korea.

RESEARCH EXPERIENCE **Thermal Energy Engineering Lab, The University of Tokyo** **Tokyo, Japan**
Research Assistant (Advisor: Prof. Junichiro Shiomi) *Sept. 2025 – Nov. 2025*
Project: “Machine learning-based thermal emitter design”

- Developing a Bayesian Optimization of Combinatorial Structures-Determinantal Point Process (BOCS-DPP) framework for pixelated 2D multiband selective thermal emitter design.
- Implemented a Python-based RCWA (S^4) to generate candidate structures and evaluate figure of merit, utilizing high-index dielectric Ge to avoid the convergence problems typical of metallic structures.

Nano Photonics Lab, KHU**Yongin, Korea***Research Assistant* (Advisor: Prof. Sun-Kyung Kim)*Jan. 2023 – Present*

Project: “Machine learning-based Antireflective (AR) Coating Design”

Collaborators: Korea Research Institute of Chemical Technology (KRICT)

- Developed a binary optimization framework based on Factorization Machines (FM) for designing deep ultraviolet (DUV) AR coatings, achieving 0.6% average reflectance on curved CaF₂ lenses.
- Built Python code for phasor analysis of optimized DUV AR coatings and broadband IR filters.

Project: “Microprism Design for Enhancing Photovoltaic (PV) Cell Efficiency”

Collaborators: Hyundai Motor Group

- Computed angle-resolved transmittance of wedge-shaped microprisms using LightTools.
- Achieved near-unity front-side transmission and 81% backward redirection toward PV cells, validated through experimental measurements.
- Conducted outdoor tests with microprism sheets mounted on thin-film bifacial perovskite PV cells, resulting in a 17% average improvement in PV power generation.

Center for Quantum Information, Korea Institute of Science & Technology**Suwon, Korea***Research Assistant* (Advisor: Dr. Hyang-Tag Lim)*Jul. 2022 – Sept. 2022*

- Formulated a Sagnac interferometer to generate high-purity entangled photon pairs via PPKTP type-II spontaneous parametric down-conversion.
- Tuned lens spacing and nonlinear crystal temperature to optimize indistinguishability, achieving 99% visibility in the Hong-Ou-Mandel dip.

HONORS AND AWARDS**Outstanding Graduate Scholarship, KHU***2024 & 2025*

- Full-tuition master’s degree scholarship awarded to top-performing Bachelor’s graduates.

Excellent Graduation Poster Award, KHU*2023***Overseas Academic Program Scholarship, KHU***2023***Academic Excellence Award and Scholarship, KHU***2022 & 2023***TEACHING EXPERIENCE****Semiconductor and Properties Processing Lab, KHU****Yongin, Korea***Teaching Assistant* for an upper-level selective course*Mar. 2025 – Jun. 2025*

- Guided advanced undergraduates in semiconductor device processing techniques (thermal/e-beam evaporation, ellipsometry, AFM, FTIR, RIE, and laser interference lithography).

TECHNICAL SKILLS**Experimental Skills:** Thermal/E-beam evaporator, Ellipsometer, FTIR, AFM**Simulation Skills:** FDTD, RCWA, COMSOL, LightTools**Programming:** MATLAB, Python**EXTRACURRICULAR ACTIVITIES****Saarland University****Saarbrücken, Germany***Overseas Academic Program* (Capstone Design)*Jul. 2023*

- Conducted a campus-wide survey to identify challenges in food storage and expiration tracking, leading to the development of an AI-driven solution that scans ingredients, estimates expiration dates, and alerts users to help reduce waste.